Serious Game Design Concept
“Life.exe”
ARIN6904

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Part 1 Game pitch

Our game is named “Life.exe, it is an independent pixel-based simulation/strategy game for PC and consoles. Players create a new save file after each playthrough, starting on the first in-game day. The game lasts 10 to 15 days, and each playthrough takes 45 to 60 minutes in the real world. The numerical system includes a player's account balance, which can be negative, a mental health value (0-100), and a physical health value (0-100). These three values are interconnected. For example, a low balance can lead to a decline in mental health, while a low physical health value can affect work efficiency and income.

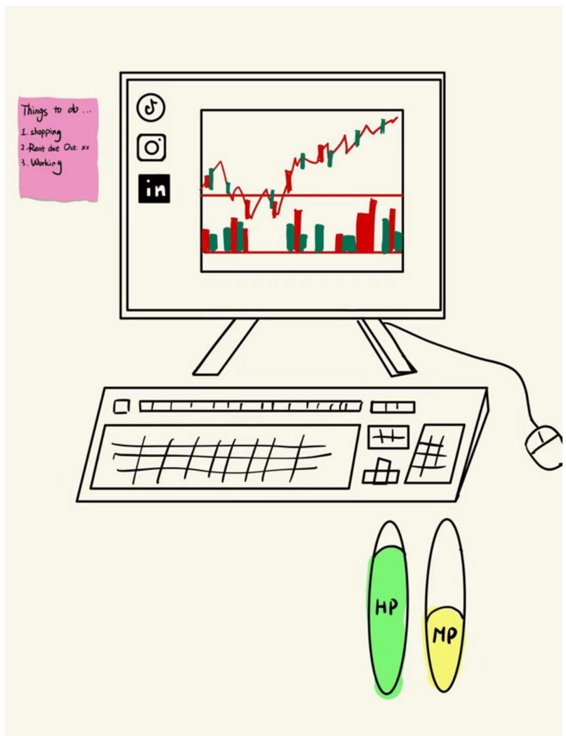
The game's main interactive scenarios include information interaction and daily life. Information interaction takes place on computer monitors, mobile phone screens, and TV screens, all displayed in a pixelated style. First, there are five or six desktop applications on the computer, which players can interact with using a mouse.

There are five applications. NewsNow (a news website) updates 2-3 news items daily, randomly mixing true and false information, such as "The stock market continues to rise, and experts predict a surge in tech stocks."

VidBox (a video website) displays entertainment content, such as funny videos and music. This content can be completely meaningless, but spending time watching it can restore a small amount of health. It may also contain information that influences player decisions.

ShopEZ (a shopping website) is the primary means of online shopping within the game. Players can purchase food, medicine, entertainment items, and other items that affect health and mental health. For example, canned food can quickly restore a small amount of health, while sleeping pills can force sleep but reduce health.

InvestPro (investment software) is a key component of the player's financial system, allowing players to purchase randomly fluctuating stocks or invest in other assets. The app's design is similar to real-world stock trading software, but it's much simpler, featuring only around 20 tradable stocks and displaying a chart showing the price of each. The app also includes a news section where players can check out trend forecasts for selected stocks. Daily news events directly affect stock prices, but players can be misled by false news and make poor choices. This is especially true for certain high-risk, high-return investments, which can either make players rich overnight or go bankrupt suddenly.



Information interaction example: PC Monitor

JobBoard (a job search website) is the only source of funding for players besides investments. On this app, players can view all available job types, including short-hour, low-paying jobs like delivery driver, salesperson, and warehouse manager; as well as long-hour, high-paying, and high-pressure jobs like programmer and office worker (long work hours can affect health and mental well-being). Wages are paid at the end of each day and automatically added to the player's account balance.

Finally, the CivicBank app is the primary place for players to check their real-time account balances. It also offers a variety of loan options, allowing users to borrow as needed. Of course, due to the short duration of the game, repayments are calculated daily. Players can also view their debt status on the app. Furthermore, the app provides a rating based on the player's financial status: healthy, average, poor, or risky, providing user-friendly reminders.

Mobile phones are much simpler than desktop apps, essentially serving as "lightweight versions" of desktop applications. The main difference is that mobile apps offer more frequent updates, but the quality of information is lower, and there's more misinformation. Players face a direct trade-off between convenience and accuracy.

NewsNow Lite updates news simultaneously, just like desktop apps, but with a streamlined layout, displaying only the headline and a summary. Players must click on the full text to see the full story.

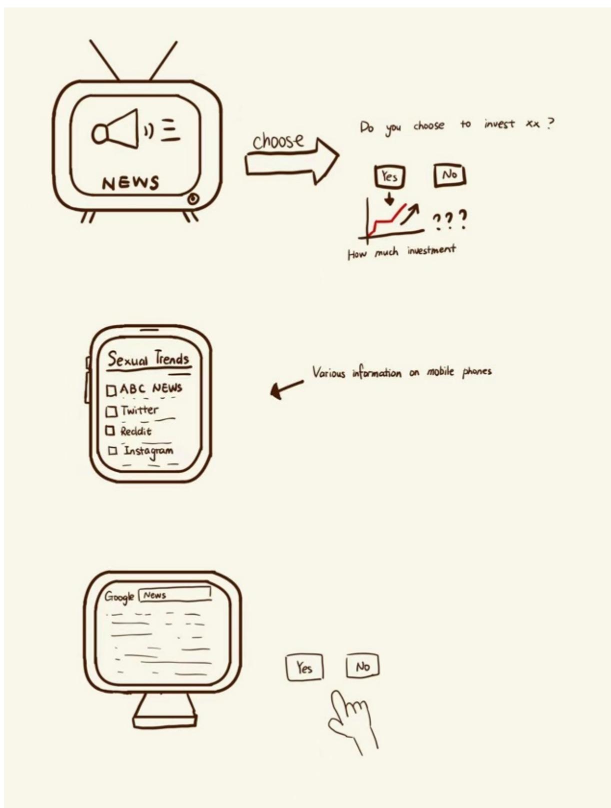
VidBox Mobile features short videos (less than 30 seconds), which can lead to addiction and lower mental health. Furthermore, occasional videos with rumors, such as "buy XX stock and you'll make money," can further influence player decisions.

InvestPro Mobile offers the same functionality as the desktop app, but its interface is simpler and lacks detailed charts, which can be misleading. Therefore, it's recommended to make decisions using the desktop app.

JobBoard Mini offers part-time jobs that allow quick delivery, such as food delivery, but these jobs generate less income than those offered through the desktop app.

Finally, Social+ is a mobile-only app that includes text messaging and calling functionality, with an interface similar to a simplified WhatsApp or WeChat app. Messages primarily consist of occasional messages from friends. For example, accepting an invitation to "go out for a drink" can restore mental health, but this also incurs additional expenses. Players may also receive scams, such as "click on a link to claim a grand prize," which can lead to significant losses if players believe them. Group chats also contain occasional gossip and investment information, though accuracy is often low.

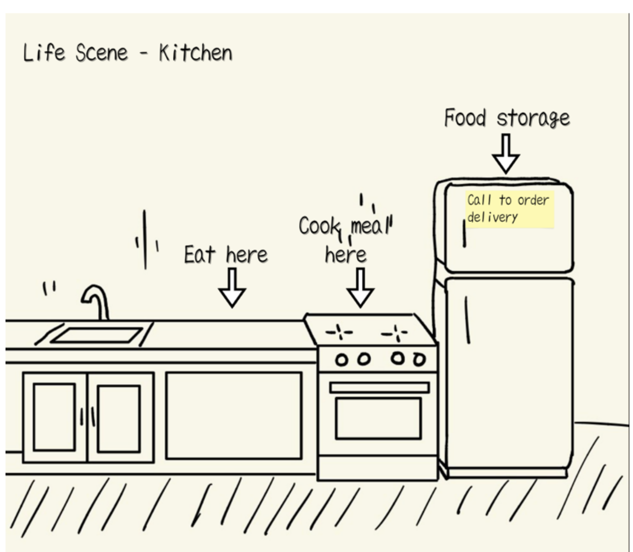
The TV scene is much simpler, allowing players to change channels with the mouse. There are five or six channels, including news channels with both left-wing and right-wing leanings, covering social, economic, and political news. The weather channel affects food prices and the difficulty of going out to work, while the entertainment channel can restore mental health. Some news items can be directly interacted with, such as "A large-scale protest has broken out. Do you decide to join?" Choosing to participate may increase mental health or increase the player's chance of being arrested and fined.



Basic game mechanism example

Life scenes include the balcony, kitchen, and bedroom. Their main function is to change the player's values or status. For example, making a phone call or smoking on the balcony, cooking or eating takeout in the kitchen, and sleeping in the bedroom can skip the fertility time for the day, or choose to stay up late to increase in-game time, but at the expense of health.

The game uses a multiple-ending system based on a combination of account balance, health value, and mental health value. It can be roughly divided into three categories: Wealth ending. If the balance is greater than 50,000, it triggers the capitalist or rich man ending. If it is less than 0, it triggers the homeless ending. For the psychological ending, if the mental health value is greater than 50, it will not affect the game ending. If it is less than 20, it will immediately trigger the psychological crisis ending or even the suicide ending. Similarly, for the physical ending, if the health value is above 50, it will not affect the game ending. If it is less than 20, it will randomly trigger sudden illnesses such as heart disease and stroke. There is also an achievement system in the game. For example, if the player does not believe any fake news, the "Distinguishing Right from Wrong" achievement will be triggered. If the player works part-time as a takeout deliveryman for more than 5 consecutive days, the "Takeout Expert" achievement will be triggered, and so on.



Life scene example: Kitchen

Once all preliminary analyses are completed, the game must then address the analysis of its “background” and “audience analysis”.

Background This project is a serious game that combines learning goals with immersive play to address four key real-world challenges: information literacy, mental health, financial decision-making, and cross-cultural adaptation. The gameplay is built around five interconnected desktop applications—NewsNow, VidBox, ShopEZ, InvestPro, and JobBoard—each targeting specific skills. A unified state model links psychological, economic, and informational variables, creating a dynamic feedback loop where actions in one area affect outcomes in others. The game runs on PC and mobile platforms, offering the same core mechanics and storyline but adapting presentation and interaction to each platform’s strengths.

Audience The primary audience consists of 18–25-year-old university and vocational college students who are currently enrolled in full-time study. Many own only a standard office-grade laptop, often provided by their institution or purchased for coursework, and do not have access to high-end gaming PCs or consoles. They typically use their devices for academic research, online classes, and light entertainment, rather than competitive or graphics-intensive games.

These students are digitally connected but time-constrained. They often study in shared spaces such as libraries, dorm rooms, or cafés, and prefer games that can be paused or resumed without losing progress. Their gaming habits lean toward narrative-driven, strategy, or simulation titles that offer intellectual engagement rather than fast-paced reflex challenges. Many are bilingual or learning a second language, and they frequently consume media from both domestic and international sources, making them receptive to cross-cultural content.

A secondary audience includes early-career professionals aged 25–30 in fields such as media, education, and business, who seek to improve their decision-making, media literacy, and cross-cultural communication skills. They often play games on mobile devices during commutes or short breaks. Both groups value practical skill development alongside entertainment and are motivated by experiences that feel relevant to real-world challenges.

Client / Relevant Organisations: Potential clients include schools, universities, and NGOs focused on digital citizenship, media literacy, and mental health. Corporate training departments could use the game for cross-cultural communication and crisis management exercises. Relevant partners might include news organisations for case studies and financial education bodies for investment scenarios.

Competitors: Comparable titles exist but differ in scope and integration. *The Most* (a Chinese real-life scenario simulation) focuses on urban survival and socio-economic pressure but remains single-platform. *Papers, Please* explores document inspection and moral choices with limited information flow. *This War of Mine* presents moral dilemmas in a war setting but does not simulate modern information ecosystems. On mobile, *Reigns* and *BitLife* offer quick decision-making but lack cross-platform data links.

Technical Feasibility: The concept is technically feasible using established cross-platform engines such as Unity or Godot. Core requirements include real-time data synchronisation across devices, adaptive UI for each platform, and modular content management for future expansion. The main challenge lies in maintaining consistent gameplay quality and minimal latency across platforms.

Precedents: While entertainment games like *Genshin Impact* and *Fortnite* have achieved cross-platform progression, few serious games combine PC analytical depth and mobile quick-decision contexts within a single unified system. This project's novelty lies in using cross-platform synergy as a deliberate teaching method, not just a convenience feature. By integrating multiple skill domains into one coherent experience, it offers a rare and transferable learning environment that mirrors the complexity of real-world decision-making.

This game combines learning goals with immersive play to address real-world challenges. It focuses on four connected areas: information literacy, mental health, financial decision-making, and cross-cultural adaptation. Players interact with five main desktop applications; each linked to a specific skill.

For example:

NewsNow delivers a mix of accurate and false reports, training players to judge credibility and improve media literacy.

VidBox streams both entertainment and subtle persuasive content, showing how media can support mental recovery but also shape bias.

ShopEZ lets players buy items that affect health and wellbeing, linking consumer choices to resource management.

InvestPro simulates stock market changes and the effect of news on investments, helping players practise risk–reward balance.

JobBoard offers jobs with different pay, hours, and stress, prompting players to weigh income against lifestyle and health.

The game's innovation lies in its multi-dimensional state model. Psychological, economic, and informational factors are connected, so a choice in one area can affect others. Information credibility changes over time, with player actions and simulated events, reflecting the unstable nature of real information systems. The design uses the strengths of each platform: PC for multi-window analysis and planning; mobile for fast, fragmented updates that increase decision pressure.

The narrative includes branching paths, real-world case adaptations, and culturally diverse situations, which increase both relevance and replay value.

Learning happens in two ways. First, players gain skills without realizing it, such as evaluating information, managing emotions, planning finances, and adapting to cultural differences. Second, the game analyses player behaviour and gives personalised feedback. Its modular design allows new scenarios and tasks to be added for different educational or research needs. By combining realism, interactivity, and adaptability—and by embedding learning into the logic of its five core applications—this project offers a unique and effective approach to serious game design, with strong potential for lasting educational impact and player engagement.

A main strength of this project is its design for different platforms, making sure each device gives a unique way to learn and play. On PC, the game works as the main analysis centre. Players can use several applications at the same time, compare data, and make long-term plans. The interface has multi-window views, sortable tables, and clear charts to support decisions based on facts. This is different from games like *The Most* (a Chinese real-life scenario simulation), where PC play focuses mainly on survival work cycles and social pressure. While *The Most* captures the grind of urban survival, our PC version formalises credibility assessment and integrates market–media feedback loops, allowing news events to directly influence asset prices, job availability, and wellbeing metrics.

By giving each platform a clear role—PC as the main place for analysis and mobile as the fast but noisy decision tool—the game helps players combine information and strategies from different angles. Players who use both platforms together can reach higher credibility scores, keep their investments more stable, and maintain better mental and physical health. This teamwork between platforms makes the project stand out from other serious games and also makes it a flexible learning tool. It can help players build skills in media literacy, financial decision-making, and cross-cultural understanding, all in ways that connect to real-life situations.

Considering the variety of devices on the market and the fact that everyone owns devices across multiple platforms, we have prepared integrated solutions for multiple platforms.

A fully interconnected architecture between the PC and mobile platforms offer significant advantages in both gameplay depth and educational impact. Both platforms deliver the same core game, with identical mechanics, narrative, and progression systems, ensuring that the player's experience remains consistent regardless of device. What changes is the form of presentation and interaction: the PC version frames the experience through a multi-window analytical interface and the mobile version condenses the same information into streamlined feeds and notifications.

Part 2 Analysis

A great game is often more than just a way to pass the time. It allows players to learn through repeated attempts. Every choice they make, every failure and success, gradually shapes their understanding of the game world and even influences their strategy for the next playthrough. This characteristic of learning through play corresponds to an important concept in game studies known as "Procedural Rhetoric" (Bogost, 2008, p.125). It emphasizes that games are not merely entertainment, but a means of expressing ideas through rules and systems. Players are not told what to think, but gradually internalize these perspectives into their own understanding through continuous manipulation, receiving feedback, and adjusting strategies. As Bogost (2008, p.121) states, "The rules do not merely create the experience of play—they also construct the meaning of the game." At the core of this theory lies the principle that rules not only dictate how players engage with a game, but also determine what they learn and experience from it. In this game, this concept is meticulously implemented in every interactive detail. The information ecosystem is divided into three primary terminals: computers, mobile phones, and televisions, each with distinct update speeds, information accuracy, and narrative biases. The mobile end is faster, but it's full of rumors and scams. The computer end is slow but reliable, and the TV end has a left-right bias. Players need to make judgments every day, whether to choose the convenient mobile push to take risks, spend time verifying on the computer to ensure investment safety, or rely on television to obtain one-sided but fast social news. Every click by the player directly impacts InvestPro's stock market fluctuations and account balance, forming a clear chain of causality. This divergence in information channels represents the most quintessential manifestation of programmatic rhetoric. Players confront the speed, quality, and bias of diverse information sources daily, deciding within a limited time whether to verify by themselves or follow the crowd. This mechanism transforms media literacy from theory into concrete behavioral challenges. It forces players to weigh efficiency against risk, experiencing losses from rumors or delayed gains from verification, thereby gradually instilling a visceral understanding that “information quality comes at a cost.” This visceral understanding proves more persuasive than directly instructing players to verify news, as it is internalized through failure and consequence.

At the same time, the three metrics, account balance, mental health, and physical condition, interact with one another, forming an interconnected dynamic system. Low balances not only cause psychological values to decline but also subject players to daily “accumulating pressure.” When psychological values drop to critical levels, anxiety events are triggered, further impairing investment judgment. If health points drop too low, the player is forced to rest immediately, losing a day's income. This chain reaction gradually makes players realize that every instance of overspending or neglecting health will lead to greater costs down the line. For example, players who stay up late to earn an extra delivery order may experience decreased stamina the next day, leading to lower efficiency, reduced income, and even accelerating their mental health toward the breaking point. The entire system acts like a mirror, constantly reflecting the consequences of players' decisions unchanged. This makes the costs of short-term overindulgence tangible and perceptible, forcing players to rethink their strategies in the next round. Gradually, the players find a rhythm that balances convenience with risk, and rewards with well-being.

This clear and immediate chain of causality embodies what Doug Church (Salen Tekinbaş & Zimmerman, 2005) termed "perceivable consequence." The game world responds distinctly to every player action through numerical fluctuations, event triggers, and interface feedback, enabling players not only to understand what has occurred but also to deduce why it happened. This design allows players to establish a psychological model of their choices and gradually understand the operating rules of the system. When they suffer investment losses due to easily believing in fake news, miss out on earning opportunities due to prolonged phone usage, or lose a day's income due to being forced to rest due to staying up late, these are all learning points for players. It is this causal transparent feedback loop that not only enhances the depth of learning, but also strengthens the replay value, as players can always find the motivation to do better next time.

The emotional curve is actually the most overlooked but crucial part of this game. Rules not only allow players to make choices and roll numbers, but also gradually stimulate their emotions, making them truly remember this experience and willing to try another round. The game's rules deliberately orchestrate emotional highs and lows, allowing players to experience pressures, anxieties, relief, and a sense of accomplishment akin to real life. For example, when a player impulsively follows the crowd in investments, causing their account balance to plummet and their psychological state to drop to the threshold, triggering anxiety events, this continuous negative feedback loop can make the player genuinely feel regret and ponder, “Should I do this again next time?" This design is far more persuasive than simple red prompts or failed subtitles, as players go through a complete emotional fluctuation process, from initial greed, to panic to rational reflection. Similarly, positive feedback is carefully placed at critical junctures. After carefully verifying the news and making sound decisions on the same day, the player. The next day, seeing the stock market rise, balance increase, and psychological value rebound, one will gain a sense of achievement and security. It is this combination of rules and feedback that turns simple numerical management into a critical thinking field. Players gradually learn to balance convenience and risk, short-term benefits and long-term health, information speed and reliability. These meanings are not conveyed through plot lines, but naturally grow through interface friction, numerical fluctuations, and cascading consequences. Therefore, the rules of this game not only create a tense and realistic gameplay experience but also provide players with a profound metaphor for modern society, allowing them to practice how to make sustainable decisions under information overload and economic pressure.

While the mechanics of this game are highly innovative and realistic, it also presents potential weaknesses and challenges. First, information overload can lead to negative gamer experiences. The game utilises six desktop applications, five mobile apps, and multiple TV channels to push notifications to players. Each platform has varying update rates and a mix of true and fake news. Although this complex information environment truly reflects the characteristics of modern digital life, it can also create cognitive overload for players. According to cognitive load theory, Kahyaoglu Erdogmus and Kurt (2024) mentioned that excessive information input can lead to cognitive overload, hindering effective learning (p.12947). Due to the learners having limited working memory capacity. Players must juggle multiple tasks within the 10-15 day game cycle, including distinguishing true from fake news, investment decisions, job selection, and health management. This multi-tasking cognitive demand may exceed the processing capabilities of average players. For the target user group, particularly those who prefer casual games, an overly complex information environment can cause frustration, leading to game abandonment or inappropriate learning outcomes.

Secondly, the oneness ending system of the game may be a disadvantage in long term gameplay. While the game offers a variety of endings, including "Rich Man," "Homeless," and "Suicide," it seemingly offers plentiful and diverse options. In fact, these endings are essentially based on different combinations of three core numerical indicators: health, financial status, and mental state. From a replayability perspective, the game lacks sufficient randomisation, hidden quests, or progressively unlockable, deeper storylines. This results in players mastering all primary game paths and ending triggers after completing a few loops. LeBlanc (2005) points out that drama as an aesthetic is based on the balance between uncertainty of outcome and necessity of process (p.357). However, the current game's highly predictable paths and oneness feedback loops make it difficult to create this kind of emotional tension and conflict. It leads to player tiredness.

Furthermore, the existing ending lacks guiding open-ended questions and extended thought mechanisms. It results in a lack of sustained attention and motivation for action after completing the game. According to the article, serious games need to be entertaining and engaging. They aim to enable players to understand, identify with, sympathise with, or remember the characters without compromising the gameplay experience (Caserman et al., 2020). From a serious games perspective, this design limits the game's potential for extended impact and cultural transmission. It makes it difficult for players to develop deep emotional resonance and reflective thinking after the game ends. Caserman et al. (2020) mentioned that serious games should provide players with feedback, including post-game feedback, which can enhance learning outcomes. Therefore, the ending lacks the factors that can generate sustained attention, critical thinking, or even promote attitudinal change and behavioural reflection on real world issues. It did not satisfy the feedback elements that serious games should include.

Another potential weakness is that the game may be limiting for players. Yee (2014) summarises player motivations for gaming into three core dimensions, which are "Achievement," "Immersion," and "Social" (p.29). However, this game primarily satisfies the achievement cluster player. It provides satisfaction through mechanics such as numerical optimisation, strategy development, and goal achievement. For players who prefer "Immersion," the game design only partially fulfils these needs. For instance, the characters lack backstories and personality traits, whereas the game fully replicates some real-life actions. For players who prefer "Social Interaction," the game lacks virtually any social elements. In the real world, people's information judgment relies heavily on social interaction, teamwork, and discussion. However, these crucial social learning mechanisms are absent in the game. The lack of social mechanics prevents players from experiencing the crucial role of collective intelligence in information sifting. It is challenging to achieve effective interaction and emotional connection between players, such as collaboratively analysing news, discussing investment strategies, and sharing experiences. The lack of competition, such as leaderboards, also weakens the game's incentives. Players are unable to compare their performance with others to gain a sense of achievement and the motivation to improve.

Furthermore, the game lacks an internal community platform to support depth communication and discussion among players. This limits educational value to the individual level and prevents it from promoting collective learning and amplifying social impact. This imbalance limits the game's audience reach and market potential, and potentially weakens its educational and entertainment value.

Therefore, the challenge for this serious game is finding a balance between educational objectives and gameplay enjoyment. According to the article, players play games to enjoy the game's aesthetics and attract emotional responses (LeBlanc, 2005, p.357). While this game incorporates rich mechanics and diverse information channels to train players' discernment, its core gameplay is essentially a simulation of real world social issues. These serious topics are complex and emotionally charged. Overemphasising educational aspects of the game would undermine its entertainment and immersive nature. Players would be required to invest more knowledge base and emotion to discern complex information, making it less engaging. In contrast, overly content with entertainment needs can prevent players from truly understanding the game's deeper meaning. As a serious game, this design concept demonstrates profound theoretical thinking and innovative design concepts. However, it also faces common challenges in the serious game field. Therefore, serious game design should balance between educational functionality and user experience. On the one hand, it ensures the effective communication of learning objectives, while on the other hand, it must preserve the game's inherent fun and appeal.

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